VALIDATION PLAN
OPERATIONAL QUALIFICATION
CORRELATION ANALYSIS MODULE

JR Validated Environment — Correlation Analysis Module v1.0

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Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-20	Joep Rous	Initial release. Covers all 4 Corr scripts in module v1.0.

Approval Signatures

Role	Name	Signature	Date
Author			
Reviewer			
Approver (QA)			

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Correlation Analysis module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each Correlation Analysis script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the Correlation Analysis module.

2. Scope

2.1 In Scope

- All 4 R scripts in the repos/corr/R/ directory
- Correct computation of correlation coefficients, regression coefficients, CI bounds, and p-values for valid inputs
- Correct rejection of invalid inputs with informative error messages
- Correct file I/O behaviour (CSV input, PNG output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup
- --xcol / --ycol column selection and --conf confidence level handling

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — method comparison studies using production data are conducted separately under individual design verification protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, grid) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Pearson, Spearman, OLS, Passing-Bablok) — method references are cited in Section 3

3. Regulatory Basis and References

3.1 Regulations

Reference	Title
21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 11	Electronic Records; Electronic Signatures

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002
FDA CSA	Computer Software Assurance for Production and Quality System Software;	September 2022

	Draft Guidance	

3.3 Standards and Methods

Reference	Title
Pearson (1895)	Pearson K. Notes on regression and inheritance in the case of two parents. Proceedings of the Royal Society of London, 58, 240–242.
Spearman (1904)	Spearman C. The proof and measurement of association between two things. American Journal of Psychology, 15(1), 72–101.
Passing & Bablok (1983)	Passing H, Bablok W. A new biometrical procedure for testing the equality of measurements from two different analytical methods. J Clin Chem Clin Biochem, 21(11), 709–720.
ISO 5725-2:2019	Accuracy (trueness and precision) of measurement methods and results — Part 2: Basic method for the determination of repeatability and reproducibility.
ISO 13485:2016	Medical devices — Quality management systems — Requirements for regulatory purposes.

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment v1.0.0
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_20260311_205146.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite v1.1.0

4. Definitions and Abbreviations

Term	Definition
Pearson r	The product-moment correlation coefficient: a measure of linear association between two continuous variables; ranges from −1 (perfect negative) to +1 (perfect positive).
Spearman ρ	The rank-based correlation coefficient: a non-parametric measure of monotonic association; does not require normality.
Fisher z	A transformation of Pearson r used to construct symmetric CIs: z = atanh(r).
OLS	Ordinary Least Squares — fits a straight line minimising the sum of squared vertical residuals; assumes the x variable is measured without error.
Passing-Bablok	A robust regression for method comparison that treats both variables symmetrically and does not require a gold-standard reference method; slope = 1 and intercept = 0 indicate interchangeable methods.
Cusum test	Cumulative sum of residual signs used to test linearity assumption in Passing-Bablok; max |CS| compared to 1.36√n at 5% significance.
CI	Confidence interval — range of plausible values for a parameter at the stated confidence level.
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
UR	User Requirement
TC	Test Case
jrrun	The validated bash entry-point script that sets the controlled environment before executing any script
CSV	Comma-separated values input file
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. System Description

5.1 Module Overview

The Correlation Analysis module is an extension of the JR Validated Environment providing four R scripts for correlation and regression analysis. Scripts are installed under repos/corr/ and are invoked via wrapper

commands (e.g., jrc_corr_pearson). The module shares the core environment infrastructure (jrrun, renv library, integrity checking, logging) with the community script suite. No additional R packages are required beyond those already installed (ggplot2, grid).

5.2 Scripts Under Test

Script	Method	Input Columns	Key Outputs
jrc_corr_pearson	Pearson product-moment correlation	x, y (configurable)	r, 95% CI (Fisher z), t-stat, p-value, scatter PNG
jrc_corr_spearman	Spearman rank correlation	x, y (configurable)	rho, S statistic, p-value, rank scatter PNG
jrc_corr_regression	Ordinary least squares	x, y (configurable)	slope, intercept, R^2, CI, F-stat, two-panel PNG
jrc_corr_passing_bablok	Passing-Bablok regression (native)	x, y (configurable)	slope + intercept with CI, proportionality test, bias test, Cusum test, PNG

5.3 Intended Use

Correlation Analysis scripts are intended for use by engineers evaluating linear and monotonic relationships between paired measurements in medical device design verification. Intended uses include:

- Quantifying the linear association between two continuous variables before or after a design change
- Assessing the monotonic relationship between a test method and a reference measurement
- Fitting a calibration line (slope, intercept) for a test system
- Comparing two measurement methods using Passing-Bablok regression, complementary to Bland-Altman agreement analysis
- Supporting method comparison records in the Design History File (DHF)

5.4 Hardware and Software Environment

Component	Specification
Operating system	macOS and Windows 10/11 (Git Bash)
R version	4.5 (as specified in admin/r_version.txt)
R packages	ggplot2, grid (versions as locked in admin/renv.lock)
Entry point	bin/jrrun
Module location	repos/corr/
Test runner	pytest in ~/.venvs/MyProject_oq/ (reused from core OQ venv)

6. Risk Assessment and Validation Level

6.1 Risk Classification

Correlation Analysis scripts are used to quantify relationships between paired measurements in design verification studies. Incorrect results could lead to erroneous conclusions about method equivalence, calibration, or association strength, which could propagate into the Design History File and affect regulatory submissions or patient safety decisions. This represents a high-consequence failure mode. Per the FDA General Principles of Software Validation (GPSV 2002), the level of validation effort is commensurate with this risk.

6.2 Risk Factors

Risk Factor	Assessment	Consequence

Incorrect Pearson r or CI calculation	Medium probability	High: incorrect conclusion about relationship strength used in DHF
Incorrect OLS slope or R^2	Medium probability	High: incorrect calibration or regression model
Incorrect Passing-Bablok slope CI (native algorithm)	Medium probability	High: incorrect method comparison conclusion
Silent failure (non-zero exit, no message)	Low probability	Medium: user unaware of failure
Script runs outside jrrun (wrong library)	Low probability	High: wrong package versions, undetected results

Strategy: Full OQ testing of all 4 scripts, covering both primary (happy-path) execution paths and boundary/error-handling paths. Numerical outputs are verified against independently computed expected ranges using synthetic test datasets with known properties.

7. Roles and Responsibilities

Role	Responsibility
Author	Writes and maintains this Validation Plan and the Correlation Analysis module scripts
Reviewer	Reviews test specifications for completeness, accuracy, and regulatory adequacy
QA Approver	Approves this plan prior to OQ execution
Test Executor	Executes the OQ test suite (repos/corr/admin_corr_oq) and documents results
QA Witness	Reviews execution evidence and signs the OQ execution report

The Test Executor and Author may be the same person. The QA Approver must be independent of the Author for this document.

8. Validation Approach

8.1 Overall Strategy

The Correlation Analysis module is validated as an extension of the core JR Validated Environment using the same IQ/OQ/PQ model. The core environment IQ is documented in JR-IQ-001. This document covers the OQ for the Correlation Analysis module only.

Phase	Description	Status
IQ	Core environment: R version, packages, jrrun, integrity checking. Corr module scripts are included in the project_integrity.sha256 file.	Completed — JR-IQ-001
OQ (this document)	Verifies correct operation of each Correlation Analysis script across the range of anticipated inputs	Planned
PQ	End-to-end performance with real measurement data; executed under individual design verification protocols	Out of scope here

8.2 OQ Execution

OQ tests are executed by the automated test runner repos/corr/admin_corr_oq, which:

1. Activates the OQ Python virtual environment (~/.venvs/MyProject_oq/)
2. Invokes pytest against all test_*.py files in repos/corr/oq/
3. For each test case, calls the Correlation Analysis script as a subprocess via jrrun
4. Captures exit code, stdout, and stderr

5. Evaluates pass/fail against the criteria defined in Section 13
6. Writes a timestamped qualification report to ~/.jrscript/MyProject/validation/

All test cases must pass for the OQ to be considered successful. A single failing test case constitutes an OQ failure and must be resolved before the module is released to users.

8.3 Sequence of Execution

7. Confirm test data files in repos/corr/oq/data/ are present and committed
8. Run admin/admin_create_hash to regenerate project integrity file
9. Obtain QA approval of this plan
10. Execute repos/corr/admin_corr_oq on the target machine
11. Review the qualification report
12. Document any deviations
13. Obtain QA sign-off on the execution evidence

9. User Requirements

The following user requirements define what the Correlation Analysis module must do from the user's perspective. Each requirement is traceable to one or more OQ test cases in the Requirements Traceability Matrix (Section 12).

9.1 Pearson Correlation

ID	Requirement
UR-CORR-001	The system shall compute the Pearson product-moment correlation coefficient r, its degrees of freedom, a Fisher-z-based CI at the specified confidence level, t-statistic, and two-sided p-value for a paired numeric dataset with $n \geq 3$.
UR-CORR-002	The system shall classify the correlation as weak ($\|r\| < 0.3$), moderate ($0.3 \leq \|r\| \leq 0.7$), or strong ($\|r\| > 0.7$) and indicate direction (positive/negative).
UR-CORR-003	The system shall save a scatter plot with best-fit line as a PNG to ~/Downloads/.

9.2 Spearman Rank Correlation

ID	Requirement
UR-CORR-004	The system shall compute the Spearman rank correlation coefficient rho, the S statistic, and the two-sided p-value for a paired numeric dataset with $n \geq 3$.
UR-CORR-005	The system shall classify the monotonic association as weak, moderate, or strong using the same |rho| thresholds as UR-CORR-002 and display a note that no CI is computed.
UR-CORR-006	The system shall save a scatter plot of ranks with best-fit line as a PNG to ~/Downloads/.

9.3 Simple Linear Regression

ID	Requirement
UR-CORR-007	The system shall fit an OLS regression of y on x and report: slope, intercept, R^2, adjusted R^2, residual SE, F-statistic, overall p-value, and CI for both slope and intercept at the specified confidence level.
UR-CORR-008	The system shall save a two-panel PNG (scatter + regression line + CI band; residuals vs fitted) to ~/Downloads/.

9.4 Passing-Bablok Regression

ID	Requirement

UR-CORR-009	The system shall compute the Passing-Bablok slope and intercept using the pairwise-median algorithm, and report CI bounds at the specified confidence level without assuming a gold-standard reference method.
UR-CORR-010	The system shall test whether the slope CI includes 1 (proportionality test) and whether the intercept CI includes 0 (systematic bias test) and report findings.
UR-CORR-011	The system shall perform the Cusum linearity test (critical value 1.36√n at 5%) and report whether linearity is rejected.
UR-CORR-012	The system shall save a scatter plot with Passing-Bablok regression line, CI bounds, and line of identity as a PNG to ~/Downloads/.

9.5 Common Requirements

ID	Requirement
UR-CORR-013	Each script shall accept optional --xcol and --ycol arguments to select columns from the input CSV; default column names are x and y.
UR-CORR-014	Each script shall verify RENV_PATHS_ROOT is set at startup and terminate with non-zero exit code and descriptive error if not set.
UR-CORR-015	Each script shall terminate with non-zero exit code and usage message when invoked with no arguments.
UR-CORR-016	Each script shall terminate with non-zero exit code and informative error when the input file is not found.
UR-CORR-017	Each script shall terminate with non-zero exit code when the input CSV contains fewer than 3 complete (non-NA) paired observations.
UR-CORR-018	Each script shall terminate with non-zero exit code when the specified column contains non-numeric data.

10. Acceptance Criteria

The OQ is considered successful when all of the following criteria are met:

- All 45 automated test cases defined in Section 13 pass with exit code 0 from pytest
- The qualification report generated by repos/corr/admin_corr_oq records 0 failures and 0 errors
- The qualification report is reviewed and signed by the QA Witness
- Any deviations are documented and dispositioned per Section 14

11. Test Data

All test data files are synthetic CSV files committed to repos/corr/oq/data/. They are version-controlled and included in the project integrity check. No real patient or production data is used.

File	Used By	Purpose
corr_linear.csv	TC-CORR-P-001..005, TC-CORR-S-001..005, TC-CORR-R-001..005	n=10, y≈2x+1, r≈0.999 strong positive
corr_negative.csv	TC-CORR-P-010, TC-CORR-S-010	n=10, r≈−0.999 strong negative
corr_uncorrelated.csv	(supplementary)	n=10, near-zero correlation
corr_method_comp.csv	TC-CORR-PB-001..006, TC-CORR-PB-011	n=9, y≈1.1x+2, method comparison
corr_small.csv	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009	n=2, triggers n<3 error
corr_nonnumeric.csv	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010	Non-numeric y column

12. Requirements Traceability Summary

Requirement	Test Cases
UR-CORR-001	TC-CORR-P-001, TC-CORR-P-002, TC-CORR-P-003, TC-CORR-P-004
UR-CORR-002	TC-CORR-P-001, TC-CORR-P-010
UR-CORR-003	TC-CORR-P-005
UR-CORR-004	TC-CORR-S-001, TC-CORR-S-002, TC-CORR-S-003
UR-CORR-005	TC-CORR-S-001, TC-CORR-S-004, TC-CORR-S-010
UR-CORR-006	TC-CORR-S-005
UR-CORR-007	TC-CORR-R-001, TC-CORR-R-002, TC-CORR-R-003, TC-CORR-R-004, TC-CORR-R-010
UR-CORR-008	TC-CORR-R-005
UR-CORR-009	TC-CORR-PB-001, TC-CORR-PB-002, TC-CORR-PB-003, TC-CORR-PB-011
UR-CORR-010	TC-CORR-PB-005
UR-CORR-011	TC-CORR-PB-004
UR-CORR-012	TC-CORR-PB-006
UR-CORR-013	TC-CORR-P-001, TC-CORR-S-001, TC-CORR-R-001, TC-CORR-PB-001
UR-CORR-014	TC-CORR-P-011, TC-CORR-S-011, TC-CORR-R-011, TC-CORR-PB-012
UR-CORR-015	TC-CORR-P-006, TC-CORR-S-006, TC-CORR-R-006, TC-CORR-PB-007
UR-CORR-016	TC-CORR-P-007, TC-CORR-S-007, TC-CORR-R-007, TC-CORR-PB-008
UR-CORR-017	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009
UR-CORR-018	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010

13. Test Cases

Each test case is executed automatically by pytest via repos/corr/admin_corr_oq. Commands shown use wrapper names; the test suite invokes scripts via jrrun.

13.1 jrc_corr_pearson — Pearson Correlation

TC-CORR-P-001 Happy path — exit 0, output sections present

Tests: UR-CORR-001, UR-CORR-013
Command: `jrc_corr_pearson corr_linear.csv`
Pass criterion: Exit 0; output contains "Pearson" and "Pearson r:".

TC-CORR-P-002 Pearson r label present

Tests: UR-CORR-001
Command: `jrc_corr_pearson corr_linear.csv`
Pass criterion: Exit 0; output contains "Pearson r:" label.

TC-CORR-P-003 CI bounds present

Tests: UR-CORR-001
Command: `jrc_corr_pearson corr_linear.csv`
Pass criterion: Exit 0; output contains CI values (look for "[" and "]").

TC-CORR-P-004 p-value present

Tests: UR-CORR-001
Command: `jrc_corr_pearson corr_linear.csv`
Pass criterion: Exit 0; output contains "p-value".

TC-CORR-P-005 PNG written to ~/Downloads/

Tests: UR-CORR-003
Command: `jrc_corr_pearson corr_linear.csv`
Pass criterion: Exit 0; file matching *_jrc_corr_pearson.png exists in ~/Downloads/ with mtime ≥ test start.

TC-CORR-P-006 No arguments → usage message

Tests: UR-CORR-015
Command: `jrc_corr_pearson`
Pass criterion: Exit ≠ 0; output contains "Usage" (case-insensitive).

TC-CORR-P-007 File not found

Tests: UR-CORR-016
Command: `jrc_corr_pearson /tmp/no_such.csv`
Pass criterion: Exit ≠ 0; output contains "not found" (case-insensitive).

TC-CORR-P-008 n < 3 → error

Tests: UR-CORR-017
Command: `jrc_corr_pearson corr_small.csv`
Rationale: corr_small.csv has n=2 pairs.
Pass criterion: Exit ≠ 0; output contains "3" (minimum).

TC-CORR-P-009 Non-numeric column → error

Tests: UR-CORR-018
Command: `jrc_corr_pearson corr_nonnumeric.csv`
Rationale: y column is non-numeric.
Pass criterion: Exit ≠ 0.

TC-CORR-P-010 Negative correlation

Tests: UR-CORR-002
Command: `jrc_corr_pearson corr_negative.csv`
Rationale: Dataset has r ≈ −0.999.
Pass criterion: Exit 0; output contains "-0." confirming negative r.

TC-CORR-P-011 Bypass protection — direct Rscript call

Tests: UR-CORR-014
Command: `Rscript repos/corr/R/jrc_corr_pearson.R corr_linear.csv (RENV_PATHS_ROOT unset)`
Rationale: Ensures script cannot be run outside jrrun.
Pass criterion: Exit ≠ 0; output contains "RENV_PATHS_ROOT".

13.2 jrc_corr_spearman — Spearman Rank Correlation

TC-CORR-S-001 Happy path — exit 0, output sections present

Tests: UR-CORR-004, UR-CORR-013
Command: `jrc_corr_spearman corr_linear.csv`
Pass criterion: Exit 0; output contains "Spearman".

TC-CORR-S-002 Spearman rho label present

Tests: UR-CORR-004
Command: `jrc_corr_spearman corr_linear.csv`
Pass criterion: Exit 0; output contains "Spearman rho:" label.

TC-CORR-S-003 p-value present

Tests: UR-CORR-004
Command: `jrc_corr_spearman corr_linear.csv`
Pass criterion: Exit 0; output contains "p-value".

TC-CORR-S-004 CI note present

Tests: UR-CORR-005
Command: `jrc_corr_spearman corr_linear.csv`

Pass criterion: Exit 0; output contains "Confidence interval" (case-insensitive).

TC-CORR-S-005 PNG written to ~/Downloads/

Tests: UR-CORR-006
Command: `jrc_corr_spearman corr_linear.csv`
Pass criterion: Exit 0; file matching *_jrc_corr_spearman.png exists in ~/Downloads/ with mtime ≥ test start.

TC-CORR-S-006 No arguments → usage message

Tests: UR-CORR-015
Command: `jrc_corr_spearman`
Pass criterion: Exit ≠ 0; output contains "Usage" (case-insensitive).

TC-CORR-S-007 File not found

Tests: UR-CORR-016
Command: `jrc_corr_spearman /tmp/no_such.csv`
Pass criterion: Exit ≠ 0; output contains "not found" (case-insensitive).

TC-CORR-S-008 n < 3 → error

Tests: UR-CORR-017
Command: `jrc_corr_spearman corr_small.csv`
Rationale: corr_small.csv has n=2 pairs.
Pass criterion: Exit ≠ 0.

TC-CORR-S-009 Non-numeric column → error

Tests: UR-CORR-018
Command: `jrc_corr_spearman corr_nonnumeric.csv`
Rationale: y column is non-numeric.
Pass criterion: Exit ≠ 0.

TC-CORR-S-010 Negative correlation

Tests: UR-CORR-005
Command: `jrc_corr_spearman corr_negative.csv`
Rationale: Dataset has r ≈ −0.999.
Pass criterion: Exit 0; output contains "-0." confirming negative rho.

TC-CORR-S-011 Bypass protection — direct Rscript call

Tests: UR-CORR-014
Command: `Rscript repos/corr/R/jrc_corr_spearman.R corr_linear.csv (RENV_PATHS_ROOT unset)`
Rationale: Ensures script cannot be run outside jrrun.
Pass criterion: Exit ≠ 0; output contains "RENV_PATHS_ROOT".

13.3 jrc_corr_regression — Simple Linear Regression

TC-CORR-R-001 Happy path — exit 0, output sections present

Tests: UR-CORR-007, UR-CORR-013
Command: `jrc_corr_regression corr_linear.csv`
Pass criterion: Exit 0; output contains "Regression".

TC-CORR-R-002 Slope label present

Tests: UR-CORR-007
Command: `jrc_corr_regression corr_linear.csv`
Pass criterion: Exit 0; output contains "Slope" label.

TC-CORR-R-003 Intercept label present

Tests: UR-CORR-007
Command: `jrc_corr_regression corr_linear.csv`
Pass criterion: Exit 0; output contains "Intercept" label.

TC-CORR-R-004 R-squared present

Tests: UR-CORR-007
Command: `jrc_corr_regression corr_linear.csv`
Pass criterion: Exit 0; output contains "R-squared".

TC-CORR-R-005 PNG written to ~/Downloads/

Tests: UR-CORR-008
Command: `jrc_corr_regression corr_linear.csv`
Pass criterion: Exit 0; file matching *_jrc_corr_regression.png exists in ~/Downloads/ with mtime ≥ test start.

TC-CORR-R-006 No arguments → usage message

Tests: UR-CORR-015
Command: `jrc_corr_regression`
Pass criterion: Exit ≠ 0; output contains "Usage" (case-insensitive).

TC-CORR-R-007 File not found

Tests: UR-CORR-016
Command: `jrc_corr_regression /tmp/no_such.csv`
Pass criterion: Exit ≠ 0; output contains "not found" (case-insensitive).

TC-CORR-R-008 n < 3 → error

Tests: UR-CORR-017
Command: `jrc_corr_regression corr_small.csv`
Rationale: corr_small.csv has n=2 pairs.
Pass criterion: Exit ≠ 0.

TC-CORR-R-009 Non-numeric column → error

Tests: UR-CORR-018
Command: `jrc_corr_regression corr_nonnumeric.csv`
Rationale: y column is non-numeric.
Pass criterion: Exit ≠ 0.

TC-CORR-R-010 Known slope value

Tests: UR-CORR-007
Command: `jrc_corr_regression corr_linear.csv`
Rationale: corr_linear.csv has y=2x+1+noise; expected slope ≈ 1.99.
Pass criterion: Exit 0; output contains "1.9" or "2.0" (slope ≈ 2.0).

TC-CORR-R-011 Bypass protection — direct Rscript call

Tests: UR-CORR-014
Command: `Rscript repos/corr/R/jrc_corr_regression.R corr_linear.csv (RENV_PATHS_ROOT unset)`
Rationale: Ensures script cannot be run outside jrrun.
Pass criterion: Exit ≠ 0; output contains "RENV_PATHS_ROOT".

13.4 jrc_corr_passing_bablok — Passing-Bablok Regression

TC-CORR-PB-001 Happy path — exit 0, output sections present

Tests: UR-CORR-009, UR-CORR-013
Command: `jrc_corr_passing_bablok corr_method_comp.csv`

Pass criterion: Exit 0; output contains "Passing-Bablok".

TC-CORR-PB-002 Slope label present

Tests: UR-CORR-009
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Pass criterion: Exit 0; output contains "Slope" label.

TC-CORR-PB-003 Intercept label present

Tests: UR-CORR-009
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Pass criterion: Exit 0; output contains "Intercept" label.

TC-CORR-PB-004 Cusum test present

Tests: UR-CORR-011
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Pass criterion: Exit 0; output contains "Cusum".

TC-CORR-PB-005 Proportionality test present

Tests: UR-CORR-010
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Pass criterion: Exit 0; output contains "slope" and "1" near each other (proportionality test).

TC-CORR-PB-006 PNG written to ~/Downloads/

Tests: UR-CORR-012
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Pass criterion: Exit 0; file matching *_jrc_corr_passing_bablok.png exists in ~/Downloads/ with mtime ≥ test start.

TC-CORR-PB-007 No arguments → usage message

Tests: UR-CORR-015
Command: `jrc_corr_passing_bablok`
Pass criterion: Exit ≠ 0; output contains "Usage" (case-insensitive).

TC-CORR-PB-008 File not found

Tests: UR-CORR-016
Command: `jrc_corr_passing_bablok /tmp/no_such.csv`
Pass criterion: Exit ≠ 0; output contains "not found" (case-insensitive).

TC-CORR-PB-009 n < 3 → error

Tests: UR-CORR-017
Command: `jrc_corr_passing_bablok corr_small.csv`
Rationale: corr_small.csv has n=2 pairs.
Pass criterion: Exit ≠ 0.

TC-CORR-PB-010 Non-numeric column → error

Tests: UR-CORR-018
Command: `jrc_corr_passing_bablok corr_nonnumeric.csv`
Rationale: y column is non-numeric.
Pass criterion: Exit ≠ 0.

TC-CORR-PB-011 Known slope value

Tests: UR-CORR-009
Command: `jrc_corr_passing_bablok corr_method_comp.csv`
Rationale: corr_method_comp.csv has y=1.1x+2+noise; expected P-B slope ≈ 1.1.
Pass criterion: Exit 0; output contains "1.1" (slope ≈ 1.1).

TC-CORR-PB-012 Bypass protection — direct Rscript call

Tests: UR-CORR-014
Command: `Rscript repos/corr/R/jrc_corr_passing_bablok.R corr_method_comp.csv (RENV_PATHS_ROOT unset)`
Rationale: Ensures script cannot be run outside jrrun.
Pass criterion: Exit ≠ 0; output contains "RENV_PATHS_ROOT".

14. Deviations

No deviations anticipated prior to execution. Deviations identified during execution will be recorded in the OQ Execution Report (JR-VR-CORR-001).

TC ID	Description	Root Cause	Resolution	Status

15. Open Items

ID	Description	Owner	Target Date	Status